

Krishna Tripathi

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Education

Indian Institute of Information Technology, Nagpur

December 2021 to May 2025

Bachelor of Technology in Computer Science and Engineering

Nagpur, India

Technical Skills

Languages: C++, Go, Java, Python, JavaScript, TypeScript, SQL

Backend: Node.js, Express.js, FastAPI, REST APIs, Microservices

Frontend: React, Next.js **AI/LLM:** LangGraph, MCP, RAG, Langfuse

Databases: YugabyteDB, PostgreSQL, MongoDB, BigQuery, pgvector

Cloud/DevOps: AWS (Cognito, Lambda, S3), Google Cloud Run, Docker, Jenkins, GitHub Actions

Professional Experience

Yugabyte

Bengaluru, India

Software Engineer I

September 2025 to Present

Software Engineer Intern

September 2024 to August 2025

- **Engineered Meko knowledge base uploads across 3 services** using React, Go, Python, and presigned S3 POST requests. Enforced 3 storage edge checks for file size, type, and content spoofing while offloading large transfers from API servers.
- **Built Meko's memory retrieval observability and promotion flow** with top-k retrieval across memory and knowledge base context. Replaced full conversation replay, reducing input token usage by **35%** for marketing and lead generation workflows.
- **Delivered AWS Cognito authentication across 3 identity providers:** Google, LinkedIn, and GitHub. Supported profile completion and identity linking while hardening invite, logout, and federated account flows.
- **Contributed AI/RAG integrations across Java, Python, and C++:** YugabyteDB embedding storage and chat memory in LangChain4j, plus a PgDistRagRetriever and supporting tserver changes for [pg.dist.rag](#). Added HNSW and hybrid search, metadata filters, and comprehensive integration tests; published [docs](#) through a merged upstream [commit](#).
- **Developed a database migration validator for 2 health signals:** schema version and dirty state. Blocked invalid migrations before merge and surfaced actionable GitHub feedback.
- **Engineered xCluster disaster recovery services in Go and UI**, optimizing DR node selection from **450ms to 120ms** and exposing C++ replication lag metrics through Prometheus and UI monitoring.
- **Automated a Go analytics pipeline across 5 systems:** Scarf, GCS, BigQuery, MadKudu, and Salesforce. Ingested Scarf data through GCS into BigQuery for transformation, lead scoring, and CRM synchronization.
- **Expanded the YB Voyager experience in yugabyted UI across multiple open-source commits**, including the end-to-end workflow [implementation](#). Delivered **3 user-facing capabilities:** assessment UX, stage status visibility, and progress analytics across **4 migration stages**, simplifying tracking without CLI output.

Zluri

Bengaluru, India

Software Engineer Intern

August 2024 to September 2024

- **Resolved 13+ production bugs** in JavaScript/Node.js and MongoDB. Built QA dashboards and a testing framework, increasing coverage by **20%** and query performance by **30%**.

Projects

EternalAgni: Agnihotra Reminder App

July 2026 to Present

JavaScript, Capacitor, Android, PWA, Service Workers, Local Notifications

[SOURCE CODE](#) | [INTERNAL TESTING](#)

- **Built a custom NOAA timing engine** with ± 1 -second accuracy. Generated sunrise and sunset schedules for **92 days (~3 months)** locally for offline use.
- **Delivered reliable reminders** with **2-to-60-minute** lead times. Added duplicate suppression, late delivery protection, and client-side scheduling.
- **Shipped releases across 2 platforms:** Android and PWA. Added GPS/IP fallbacks, service worker recovery, and automated APK/AAB builds.

YB Voyager Copilot: AI Powered Database Migration Assistant

January 2025 to Present

LangGraph, FastAPI, Next.js, Python, MCP, Docker, YugabyteDB, RAG

[DOCKER HUB](#) | [ARCHITECTURE](#) | [DEMO](#)

- **Architected a 3-agent LangGraph and MCP workflow** covering all 4 migration stages with grounded product documentation.
- **Built a 3-tier RAG backend and containerized 5 services** with FastAPI, pgvector, YugabyteDB, and Docker. Enabled persistent data across the workflow.
- **Automated 2 report workflows:** delivery and migration. Used GitHub Actions and Playwright to generate production sizing recommendations.

Achievements

- Received **Best Use of AI** recognition for YB Voyager Copilot at the YugabyteDB internal hackathon.
- Earned **Guardian status with a maximum rating of 2145** on LeetCode. Reached **4-star status with a 1944 rating** on CodeChef and **Specialist status with a maximum rating of 1484** on Codeforces.
- Secured **3rd place** in Algorithmia at Tantrafiesta 2023.